SEO Audit Engine — API Integration Guide

Complete Blueprint for Connecting GSC, GA4 & Semrush • Generated: October 27, 2026 • 10:47 AM IST

Executive Summary: Is guide ki madad se aap apne **SEO Audit Engine** me real-time traffic, search queries, organic rankings, CTR, bounce rate aur backlinks data ko automatically pull kar sakte hain. Service Account architecture ke through bina baar-baar login kiye backend automated reports generate karega.

1. Google Cloud Service Account Setup (Unified Key)

Service Account ek automated bot identity hai jo backend scripts ko Google APIs secure access deti hai:

1. **Google Cloud Console** (console.cloud.google.com) par jayein aur Project select/create karein.
2. **APIs & Services > Library** me jaakar **Google Search Console API** aur **Google Analytics Data API** dono ko **Enable** karein.
3. **Credentials > Create Credentials > Service Account** par click karein. Name: **seo-engine-bot**.
4. Service Account banne ke baad uspar click karein > **Keys Tab** > **Add Key > Create new key (JSON)**.
5. Ek JSON file download hogi (e.g. **credentials.json**). Iska email address note kar lein (e.g. seo-engine-bot@project-id.iam.gserviceaccount.com).

2. Google Search Console (GSC) Integration

Service Account email ko GSC Property me add karke real search clicks aur query rankings fetch ki jati hain:

1. **Google Search Console** (search.google.com/search-console) open karein.
2. Left sidebar se apni website property (e.g. https://gurupunvaanii.com/) select karein.
3. Left menu me **Settings > Users and permissions** par jayein.
4. **Add User** click karein, Service Account Email paste karein, Permission me **Full** ya **Restricted (Read-Only)** select karke add karein.
5. **Data Capabilities:** Total Clicks, Search Impressions, CTR, Average Position, Top 1000 Queries, URL Indexing Status.

3. Google Analytics 4 (GA4) Integration

GA4 Data API se real traffic sources, user behavior aur conversions capture hoti hain:

1. **Google Analytics 4** (analytics.google.com) open karein.
2. Bottom-left **Admin (Gear Icon) > Property Settings > Property Access Management** me jayein.
3. **+ Icon > Add users** par click karein aur wahi Service Account Email daal kar **Viewer** permission dein.
4. **Property ID:** Admin > Property Settings > **Property Details** me jakar 9-digit numeric **Property ID** copy karein.
5. **Data Capabilities:** Active Users, Sessions, Engagement Rate, Bounce Rate, Traffic Channels (Organic/Direct/Paid/Referral).

4. Semrush Integration (API Key / CSV Importer)

Semrush se Domain Authority, Organic Competitors aur Backlink profile pull karne ke do tarike hain:

Option A (Semrush API Key): Semrush account > Subscription > API Key copy karke direct automated sync.

Option B (CSV / Export Sync): Semrush Domain Overview & Keyword export file ko dashboard me drag-drop ya backend sync karna.

Data Capabilities: Authority Score (AS), Referring Domains, Follow vs NoFollow Backlinks, Keyword Difficulty (KD%).

5. Data Pipeline & Python Libraries

Source	Python Library	Key Metrics Extracted
Google Search Console	`google-api-python-client` `google-auth`	Clicks, Impressions, CTR, Position, Top Search Queries, Landing Pages
Google Analytics 4	`google-analytics-data`	Active Users, Sessions, Channels (Organic, Social, Direct), Bounce Rate
Semrush / SEO	`requests / JSON parser`	Domain Authority Score, Total Backlinks, Referring Domains, Organic Keywords

■ **Next Action to Connect in this Project:**
1. Apni downloaded **service_account.json** file ko project folder me save karein.
2. Apna **GSC Site URL** (e.g. *https://gurupunvaanii.com/* ya *sc-domain:gurupunvaanii.com*) aur **GA4 Property ID** provide karein.
3. Hum live collector script run karke dashboard me live traffic graphs aur tables render kar denge!